

PROFESSIONAL SUMMARY

Computer Science graduate (BS CS, DLSU-D 2021–2025) building applied AI workflows, multi-agent systems, and full-stack web applications. Experienced in multi-provider LLM orchestration (Groq, Gemini, optional local Ollama), voice pipelines, hybrid retrieval, and responsive interfaces with React and TypeScript.

PROFESSIONAL WORK EXPERIENCE

UI/UX & Frontend Engineer Intern · ROC.PH Digital Marketing Services

June 2024 – August 2024 · Cavite, PH

Designed and prototyped responsive web and mobile interfaces in Figma using design token standards and usability heuristics. Translated specifications into accessible, semantic HTML5, CSS3/Tailwind, and React components across production client applications within agile sprints.

KEY TECHNICAL IMPACT

- Engineered a bilingual voice pipeline: FFmpeg 16 kHz normalization, Groq Whisper transcription with an OpenAI fallback, and Edge-TTS voices through a custom FFmpeg filter chain.
- Architected 5 asyncio Discord agents with a Groq-to-Gemini model fallback chain, hybrid RAG (Gemini embeddings + BM25, fused with RRF), and Gmail triage across 10 labels.
- Built zero-dependency episodic memory retrieval in the Python standard library using Okapi BM25 ranking (0.8 ms median recall, 12 unit tests).
- Developed responsive, accessible web components in React, TypeScript, and modern CSS during commercial frontend internship at ROC.PH.

FEATURED ENGINEERING PROJECTS

Manas: Ciel

Voice AI Copilot & Real-Time Cognitive HUD · Solo Project

Built a bilingual JA/EN voice copilot with a custom FFmpeg voice filter, live weather (wttr.in), web search (DDGS), Obsidian vault tools, and a WebSocket HUD.

LSFM AI HQ

Autonomous Multi-Agent Operations System · Solo Project

Architected 5 Discord agent services with Groq + Gemini routing (optional local Ollama mode), a headless-Edge ATS resume compiler, and OAuth2 inbox triage.

Patriot

Community Matchmaking Mobile App · DLSU-D Capstone (Lead Mobile Developer)

Engineered real-time mobile app in React Native/TypeScript with AWS Lambda, DynamoDB, Redis caching for session state, and multi-criteria matching.

Cognitive Memory Core

Episodic Memory Retrieval Engine · Solo Project

Engineered zero-dependency episodic memory retrieval in Python standard library using BM25 keyword relevance scoring for pre-flight task context.

EDUCATION

Bachelor of Science in Computer Science (BS CS)

2021 – 2025

De La Salle University–Dasmariñas (DLSU-D)

Distributed Systems · Human-Computer Interaction · Algorithms & Complexities

TECHNICAL PROFICIENCIES & CAPABILITIES

AI & LLM Orchestration:	Multi-Agent Workflows, Hybrid RAG (Dense + BM25, RRF), Voice Pipelines (Whisper, Edge-TTS, FFmpeg), Cloud LLMs (Groq, Google Gemini), Local Inference (Ollama), Tool Calling, Structured JSON
Languages & Backend:	Python 3.11 (asyncio, standard library), TypeScript, JavaScript (ES6+), Node.js, Express, aiohttp, REST APIs, WebSocket, OAuth2, Headless Chromium, Git/GitHub
Frontend & Design Systems:	React.js, React Native, HTML5, Modern CSS3, Design Tokens, Web Audio API, Canvas 2D, Tailwind CSS, Responsive Web Design, WCAG 2.1 AA Compliance, Figma